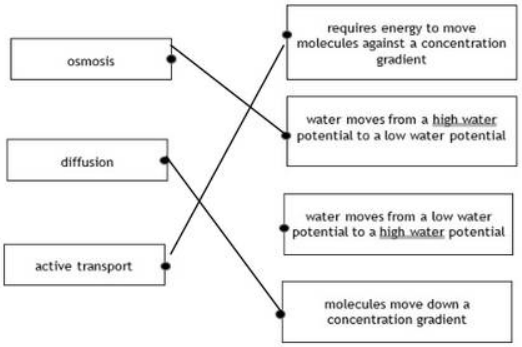
Type MS · Define specialised cells, tissues and organs and identify human tissues

Answer 1 of 11 · Question 1 part a.iii · 2 marks

2022 JUN · Paper 01 (regional R) · extract 1 of 2 · find part a.iii in this mark scheme extract

Question number				
1	(a) (i)	X = cell membrane; Y = nucleus; Z = cytoplasm;	Allow nuclear membrane	3
	(ii)	mitochondria/ribosomes/endoplasmic reticulum		1
	(iii)	• similar cells work together to form tissues; • (different) tissues work together to form organs;		2
	(b)		Reject more than one line from each process	3
	(c) (i)	A = squamous (epithelium) B = ciliated (epithelium)		2
	(ii)	60 x 40; 2400 (µm) (÷1000); 2.4 (mm);	Ecf from mp1. Max 2 marks e.g. 60 ÷ 40 (÷1000) = 0.0015	3

Total 14 marks

Type MS · Define specialised cells, tissues and organs and identify human tissues

Answer 1 of 11 · Question 1 part a.iii · 2 marks

2022 JUN · Paper 01 (regional R) · extract 2 of 2 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Define specialised cells, tissues and organs and identify human tissues

Answer 2 of 11 · Question 2 part b.i · 3 marks

2023 NOV · Paper 02 · extract 1 of 2 · find part b.i in this mark scheme extract

2 (a)		• irregular shaped cell; • nucleus labelled; • cytoplasm labelled; • cell/plasma membrane labelled;		4
(b) (i)		• similar cells/cells with similar characteristics/shape; • grouped/work together; • to carry out a particular function;		3
	(ii)	squamous epithelium;		1
	(iii)	any two from • nervous; • ciliated epithelium • blood; • bone; • (smooth/cardiac/skeletal) muscle; • connective; • skin	reject 'a' bone' reject 'a' muscle'	2

Total 10 marks

Type MS · Define specialised cells, tissues and organs and identify human tissues

Answer 2 of 11 · Question 2 part b.i · 3 marks

2023 NOV · Paper 02 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Define specialised cells, tissues and organs and identify human tissues

Answer 3 of 11 · Question 3 part a · 2 marks

2023 JUN · Paper 02 (regional R) · find part a in this mark scheme extract

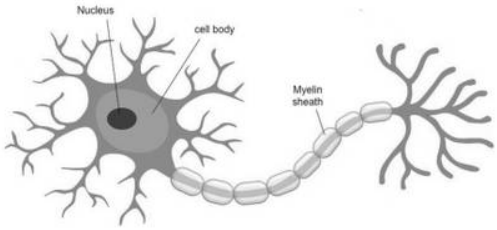
3 (a)	• contains particular features / characteristics/ adaptations; • to perform a specific function;		2
(b) (i)	• 84 ÷ 0.006; • 14000x	Allow correct final answer from a calculation that includes 84 and 0.006, max 1 mark	2
(ii)	half the amount/haploid/23 chromosomes;		1
(iii)	• to provide energy; • for swimming/movement/travelling;		2
(iv)	• 180 ÷ 0.070; • 2571; • (÷ 60) 42.85;	Allow any figure that rounds to 43 Allow ecf for third mark - an incorrect value for a calculation carried out for mps1 and 2, divided by 60 **and** a correct answer to that calculation	3 exp
(c)	Any two from: • greater magnification; • higher resolution/can see objects more clearly/in more detail; • can see a greater number of structures;		2 exp

Total Question 3 = 12 marks

Type MS · Define specialised cells, tissues and organs and identify human tissues

Answer 4 of 11 · Question 7 part b.i · 2 marks

2024 JUN · Paper 01 (regional R) · find part b.i in this mark scheme extract

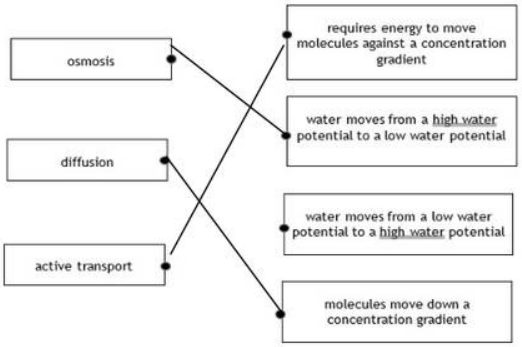
7 (a) (i)	receptor (cell)		(1)
(ii)	sensory neurone		(1)
(b) (i)	a cell that has particular features/adaptations/structures; to carry out a certain/specific function in the body;		(2)
(ii)	diagram; to include cell body at one end, with dendrites and an elongated axon twice the length of the cell body and axon ending; nucleus; cell body; myelin sheath;  	Label marks can be awarded if they draw a sensory or relay neurone	(4)
(iii)	transmits electrical impulses from the brain/CNS/spinal cord/ relay neurone; to effecter organs/muscle/gland;		(2)
(c)	100/4.0 x 10^{6}; = 0.000025 (s); = 0.025 (ms); = 2.5 x 10^{-2};	ecf for mps 2,3 and 4 Allow full marks for correct final answer	(4)

Total for Question 7 = 14 marks

Type MS · Define specialised cells, tissues and organs and identify human tissues

Answer 5 of 11 · Question 1 part c.i · 2 marks

2022 JUN · Paper 01 (regional R) · extract 1 of 2 · find part c.i in this mark scheme extract

1 (a) (i)	X = cell membrane; Y = nucleus; Z = cytoplasm;	Allow nuclear membrane	3
(ii)	mitochondria/ribosomes/endoplasmic reticulum		1
(iii)	• similar cells work together to form tissues; • (different) tissues work together to form organs;		2
(b)		Reject more than one line from each process	3
(c) (i)	A = squamous (epithelium) B = ciliated (epithelium)		2
(ii)	60 x 40; 2400 (µm) (÷1000); 2.4 (mm);	Ecf from mp1. Max 2 marks e.g. 60 ÷ 40 (÷1000) = 0.0015	3

Total 14 marks

Type MS · Define specialised cells, tissues and organs and identify human tissues

Answer 5 of 11 · Question 1 part c.i · 2 marks

2022 JUN · Paper 01 (regional R) · extract 2 of 2 · find part c.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Define specialised cells, tissues and organs and identify human tissues

Answer 6 of 11 · Question 6 part c · 3 marks

2023 JUN · Paper 01 (regional R) · find part c in this mark scheme extract

6 (a)	• general shape with nucleus at centre; • nucleus labelled; • cytoplasm labelled; • cell membrane labelled;		4
(b)	• scrape/swab cells from (inside of) cheek • smear (thinly) over slide;/place sample on slide; • add stain/cover with coverslip;		3
(c)	• cheek cell shape irregular, trachea cell oblong; • cilia present in trachea, absent in cheek cell; • position of nucleus different;		3

Total 10 marks

Question number	Answer	Notes	Marks

Type MS · Define specialised cells, tissues and organs and identify human tissues

Answer 7 of 11 · Question 2 part b.ii · 1 mark

2023 NOV · Paper 02 · extract 1 of 2 · find part b.ii in this mark scheme extract

2 (a)		• irregular shaped cell; • nucleus labelled; • cytoplasm labelled; • cell/plasma membrane labelled;		4
(b) (i)		• similar cells/cells with similar characteristics/shape; • grouped/work together; • to carry out a particular function;		3
	(ii)	squamous epithelium;		1
	(iii)	any two from • nervous; • ciliated epithelium • blood; • bone; • (smooth/cardiac/skeletal) muscle; • connective; • skin	reject 'a' bone' reject 'a' muscle'	2

Total 10 marks

Type MS · Define specialised cells, tissues and organs and identify human tissues

Answer 7 of 11 · Question 2 part b.ii · 1 mark

2023 NOV · Paper 02 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Define specialised cells, tissues and organs and identify human tissues

Answer 8 of 11 · Question 2 part b.iii · 2 marks

2023 NOV · Paper 02 · extract 1 of 2 · find part b.iii in this mark scheme extract

2 (a)		• irregular shaped cell; • nucleus labelled; • cytoplasm labelled; • cell/plasma membrane labelled;		4
(b) (i)		• similar cells/cells with similar characteristics/shape; • grouped/work together; • to carry out a particular function;		3
	(ii)	squamous epithelium;		1
	(iii)	any two from • nervous; • ciliated epithelium • blood; • bone; • (smooth/cardiac/skeletal) muscle; • connective; • skin	reject 'a' bone' reject 'a' muscle'	2

Total 10 marks

Type MS · Define specialised cells, tissues and organs and identify human tissues

Answer 8 of 11 · Question 2 part b.iii · 2 marks

2023 NOV · Paper 02 · extract 2 of 2 · find part b.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Define specialised cells, tissues and organs and identify human tissues

Answer 9 of 11 · Question 5 part a · 1 mark

2024 JUN · Paper 02 (regional R) · find part a in this mark scheme extract

5	(a)	(cardiac) muscle(s);		(1)
	(b)	• pumps blood twice in one circuit; • right side pumps deoxygenated blood to lungs; • left side pumps oxygenated blood to body;		(3)
	(c)	**C (lymphocytes and phagocytes)** The following are incorrect because: A platelets are not cells B antibodies and hormones are not cells D platelets are not cells		(1)
	(d) (i)	hormones;		(1)
	(ii)	platelets/fibrinogen/prothrombin;		(1)
	(e)	(amount of blood in 1kg) 5.5 ÷ 70 = 0.07857; (amount of blood in 95kg) 0.07857 x 95 = 7.464; (number of red blood cells in 95kg) = 7.464 x 5.9 x 10^{12} = 4.4 x 10^{13};	allow 0.08 allow 7.5 allow 4.4038 ×10^{13} allow 4.425 × 10^{13} allow 4.484 × 10^{13} full marks for correct final answer	(3)

Total for Question 5 = 10 marks

Type MS · Define specialised cells, tissues and organs and identify human tissues

Answer 10 of 11 · Question 6 part a.i · 6 marks

2024 NOV · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

<table>
<tr>
<td>6 (a) (i)</td>
<td>(cardiac muscle)
• striated/striped;
• branched;
• single/central nucleus;

(voluntary muscle)
• striated/striped;
• unbranched;
• many nuclei;

(involuntary muscle)
• non-striated/striped;
• unbranched;
• single/central nucleus;
• tapered at end;</td>
<td></td>
<td>max of 2 for each muscle

6</td>
</tr>
<tr>
<td>(ii)</td>
<td>small intestine/digestive tract/oesophagus/liver/uterus/arteries/veins/iris/sphincter muscles;</td>
<td>allow any named part of digestive tract</td>
<td>1</td>
</tr>
<tr>
<td>(b) (i)</td>
<td>1540;
(÷ 20) 77;</td>
<td>ECF incorrect addition</td>
<td>2</td>
</tr>
<tr>
<td>(ii)</td>
<td><table><tr><th>Percentage muscle mass (%)</th><th>Tally/frequency</th></tr><tr><td>70</td><td>ll 2</td></tr><tr><td>72</td><td>ll 2</td></tr><tr><td>74</td><td>llll 4</td></tr><tr><td>75</td><td>II 2</td></tr><tr><td>78</td><td>III 3</td></tr><tr><td>82</td><td>𝍸 l 6</td></tr><tr><td>84</td><td>l 1</td></tr></table></td>
<td>1 mark for suitable table, 1 mark for column headings and 1 mark for correctly inputted data and tally
allow just numbers in second column
accept just percentage/% as column heading</td>
<td>3</td>
</tr>
<tr>
<td>(iii)</td>
<td>82;</td>
<td>ECF from (ii)</td>
<td>1</td>
</tr>
<tr>
<td>(c)</td>
<td>• biceps contract;
• triceps relax;
• antagonistic pairs/work antagonistically;
• lower arm/radius pulled upwards;</td>
<td></td>
<td>4</td>
</tr>
</table>

Total Marks 17

Type MS · Define specialised cells, tissues and organs and identify human tissues

Answer 10 of 11 · Question 6 part a.i · 6 marks

2024 NOV · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Define specialised cells, tissues and organs and identify human tissues

Answer 11 of 11 · Question 6 part a.ii · 1 mark

2024 NOV · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

<table>
<tr><td>6 (a) (i)</td><td>(cardiac muscle)
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• single/central nucleus;
(voluntary muscle)
• striated/striped;
• unbranched;
• many nuclei;
(involuntary muscle)
• non-striated/striped;
• unbranched;
• single/central nucleus;
• tapered at end;</td><td></td><td>max of 2 for each muscle
6</td></tr>
<tr><td></td><td>(ii)</td><td>small intestine/digestive tract/oesophagus/liver/uterus/arteries/veins/iris/sphincter muscles;</td><td>allow any named part of digestive tract</td><td>1</td></tr>
<tr><td>(b)</td><td>(i)</td><td>1540;
(÷ 20) 77;</td><td>ECF incorrect addition</td><td>2</td></tr>
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Total Marks 17

Type MS · Define specialised cells, tissues and organs and identify human tissues

Answer 11 of 11 · Question 6 part a.ii · 1 mark

2024 NOV · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks